

NATHAN BUTLER

Data Science · Analytics · Research

nbutler428@gmail.com | (541) 520-7966 | Eugene, OR | [linkedin.com/in/nathan-r-butler/](https://www.linkedin.com/in/nathan-r-butler/) | nbutler428.github.io

PROFILE

Data science graduate student with hands-on experience building analytical pipelines, applying statistical modeling, and translating complex datasets into clear, actionable insights. Comfortable working across the full analytics workflow — from raw data ingestion and cleaning to visualization and stakeholder reporting. Passionate about using data to solve meaningful real-world problems across research, operations, and public domains.

CORE COMPETENCIES

- Python (pandas, scikit-learn)
- R (tidyverse, ggplot2)
- SQL (PostgreSQL)
- Tableau | PowerBI | Grafana
- Statistical Modeling
- Machine Learning
- ETL Pipeline Design
- Survival Analysis
- Data Visualization
- Dashboard Development
- Hypothesis Testing
- Research & Reporting

SELECTED PROJECTS

Cancer Survivability — Cox Regression Analysis

Research · 2024

Python | lifelines | Cox PH | Kaplan-Meier | SEER Data

- Applied Cox Proportional Hazards model to 10 million SEER patient records to predict time-to-event mortality
- Identified significant predictors by cancer stage and sex; achieved concordance index of ~0.85
- Built Kaplan-Meier survival curves and visualizations communicating findings to non-technical audiences

International Shipping & Ocean Nitrate Levels

Capstone · 2025

Python | K-Means | OLS | Random Forest | ERDDAP API | Spatial CV

- Investigated whether major shipping lanes measurably alter ocean nitrate concentrations across the North Atlantic
- Combined four large public datasets (NASA MODIS, NOAA OISST, World Bank, World Ocean Database) in a multi-method pipeline
- Applied K-Means clustering, OLS regression, Random Forest, and spatial cross-validation to test hypotheses

Real-Time Transit Data Pipeline

Data Engineering · 2025

Python | PostgreSQL | Apache Airflow | Grafana | ETL

- Designed and maintained a real-time ETL pipeline ingesting simulated Prague metro data into a PostgreSQL warehouse
- Automated anomaly detection across station-level events using Apache Airflow DAGs
- Built live Grafana dashboards for operational monitoring in a production-like environment

EXPERIENCE

Research Intern

Jun 2022 – Aug 2022

Monell Chemical Senses Center | Philadelphia, PA

- Designed and managed a clinical trial investigating links between Paxlovid and taste disturbance (dysgeusia)
- Built and maintained a structured trial results database; performed statistical analysis on subject outcomes
- Presented data-driven findings to stakeholders with varied technical backgrounds

Statistical Consultant

Aug 2022 – Dec 2022

Juniata College | Huntingdon, PA

- Applied ANOVA, Chi-Square, and Kruskal-Wallis tests to uncover alumni donation trends by sport and timing
- Built visualizations and dashboards enabling the development office to reduce mailing costs through donor targeting
- Delivered comprehensive written and oral reports to faculty and department stakeholders

IT Help Desk Technician

Jan 2023 – May 2024

Juniata College | Huntingdon, PA

- Diagnosed and resolved technical issues across Windows, Mac, and Linux systems for students and staff
- Maintained accurate ticket records and communicated resolution status across departments

EDUCATION

B.S. Data Science, Major in Political Science

May 2024

Juniata College | Huntingdon, PA

M.S. Data Science

August 2025

Willamette University | Salem, OR